

"ALIEN" SPECIAL SHAPE - SPECIFICATION SUMMARY

2,610 m3 / 92,171 cu ft - full-size master patterns (no scaling) - mouth O3,988 mm / hole O5,000 mm / overlap 360 mm

KEY MEASUREMENTS	METRIC	ENGLISH	
Total Volume	2,610 m3	92,171 cu ft	full-size master patterns
Scale factor (linear)	x 1.0000	1:1 - no scaling	applies to every envelope length
Envelope Height (inflated)	18,338 mm	60 ft 2 in	total ~20,620 mm / 67 ft 8 in incl. basket
Envelope Width (inflated)	17,887 mm	58 ft 8 in	
Parachute Confluence Lines (cone)	3,600 mm	11 ft 10 in	x 20
Parachute Centering Lines	3,598 - 3,639 mm	11 ft 10 in - 11 ft 11 in	start longest; trim on first inflation
Mouth Diameter	3,988 mm	13 ft 1 in	13 ft nominal
Parachute Hole Diameter	5,000 mm	16 ft 5 in	16 ft nominal
Parachute Hole Radius	2,500 mm	8 ft 2 in	
Parachute seal overlap	360 mm	14.2 in	radial, all round
Crown Line	31.0 m	101 ft 8 in	+3 m for knots/handling
Parachute red line	34.0 m	111 ft 7 in	+3 m; via apex pulley to burner frame

CONFIGURATION

Basis	master pattern set	full-size master patterns (no scaling)
Gores / panel rows	20 x 17 (Nomex + A-P) + parachute	gore length 22.83 m / 74 ft 11 in
Seams	25 mm (1 in) allowance	5/8 in double French fell; 1 in turn-backs
Hole hems + appendages	Class 3 1/2 in (12.7 mm) load tape	every panel hole, the eye rim and the antenna base - ~72 m incl. 10 %
Fabric	~748 m of 60 in ripstop (+10% = ~822 m)	Nomex ~26 m; tape ~655 m + lightweight ~242 m
Eye pods	gores 18-20 / 2-4, rows F-J; ~1.3 m proud	2 layers: envelope + Class 3 1/2 in tape + black skin (+226 mm)
Mouth applique	row C, gores 20/1/2	black crescent, 3 per-panel pieces
Antennae x2	3.75 m, row N gores 18 & 4; base O875 mm	35 deg lean; base reinforced by a 2nd panel N, not a patch
Turning vents	seams 15/16 & 16/17, rows i-J	overlap type, vent-marked i/J panels
Cloth width	1,524 mm / 60 in bolt	no cut piece in this pack is wider than 1,474 mm
Cutting table	3,658 mm / 12 ft	no cut piece is longer than 3,600 mm
Panel orientation	ROW HEIGHT across the bolt	gore width along the roll; rows 1,357-1,411 mm
Eye pod piecing	14 skin pieces per pod, 4 strips	each piece prints 1:1; feed holes cut in the envelope only

Line lengths measure to theoretical seam centers / edges - subtract loop & quick-link lengths; add 6-10 in per parachute-line end for knots (pattern-set notes).